

Customer Journey Map

2. REQUIREMENT ANALYSIS

Date	4 July 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well Being
Maximum Marks	3 Marks

Solution Requirements Document

This document lists the system requirements (both functional and non-functional) for our HDI Predictor web application.

1. Functional Requirements

FR-1: Authentication & Access

- The system **must** restrict dashboard access to authenticated users.
- Unauthenticated requests **must** be redirected to /login.
- Standard pre-filled credentials (admin / admin123) **must** be supported for easy demo purposes.
- A logout route /logout **must** clear active sessions.

FR-2: Machine Learning Prediction

- The system **must** accept four development indicators:

- Life Expectancy (20.0 to 100.0 years)
- Expected Years of Schooling (0.0 to 25.0 years)
- Mean Years of Schooling (0.0 to 20.0 years)
- Gross National Income (GNI) Per Capita (\$100 to \$150,000 PPP)
- The backend **must** validate bounds, data types, and logical conflicts (e.g. Mean Schooling > Expected Schooling) before prediction.
- The system **must** output an estimated HDI score between \$0.000\$ and \$1.000\$ and classify it into standard UN tiers (Very High, High, Medium, Low).

FR-3: User Interface & Dashboards

- The dashboard **must** display summary metrics of the training dataset.
- The frontend **must** synchronize numeric form inputs with range sliders.
- Selecting a country from the dropdown **must** trigger an API query and prefill the form.
- Tabbed components **must** display EDA visualizations (Correlation matrix, distributions, income scatter).

FR-4: History Cache & Exporters

- Prediction queries **must** save in local browser storage (`localStorage`).
- The dashboard **must** render a table showing prediction history.
- The system **must** export prediction history as a CSV file and download individual results as JSON.

🛡️ 2. Non-Functional Requirements

NFR-1: Responsiveness & UX

- The UI **must** be fully responsive, scaling properly from desktop down to mobile viewports (360px).
- The style **must** support light and dark modes, saving preference in local cache.
- Interface **must** follow modern glassmorphic cues (blur filters, soft borders, radial gradients).

NFR-2: Performance & Scalability

- The page load time **should** be under 1.5 seconds.
- Predict API response latency **must** be under 100ms .
- The model training pipeline **must** support automated fallback to high-quality synthetic generation if external UNDP downloads fail.

NFR-3: Testability

- The codebase **must** achieve a high coverage test suite checking ML predictions, validations, and server route codes.